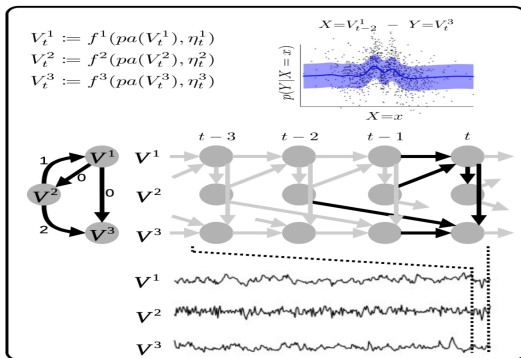


Causality for Time Series

Challenges

A d-variate time series X defined on discrete-time, for a fixed t , each X_t is a vector $(X_t^{(1)}, \dots, X_t^{(d)})$, where $X_t^{(p)}$ is the measurement of the p th time series at time t .

a Observational SCM



Question : extend causality to time series?

Challenges

- Why is causality for time series so natural?
- An underlying assumption : CAUSE is before EFFECT

Challenges

Do we take into account instantaneous effects or not?

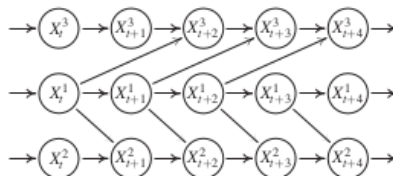


Figure 10.1: Example of a time series with no instantaneous effects.

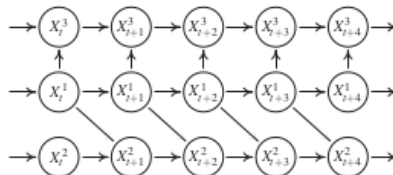


Figure 10.2: Example of a time series with instantaneous effects.

Extracted from Ch 10 of Peters et al 2017

Challenges

What temporal sampling for observational data ?

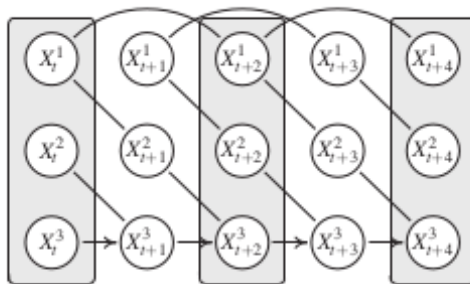


Figure 10.4: Example of a subsampled time series: only the variables in the shaded areas are observed.

Extracted from Ch 10 of Peters et al 2017

Background

Extension of Structural Causal Model to time series

Model

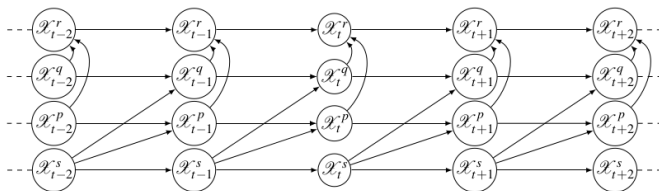
$$\forall t, \forall p, X_t^{(p)} \leftarrow f(Pa(X_t^{(p)}), \xi_t^{(p)})$$

where

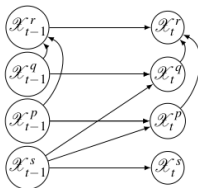
- f a real-valued multivariate function
- $\xi_t^{(p)}$ some noise serially and mutually independent of each other and independent from all the causes of $X_t^{(p)}$

Background

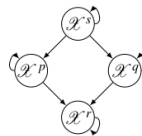
Different time causal graphs



Full Time Causal Graph (a)



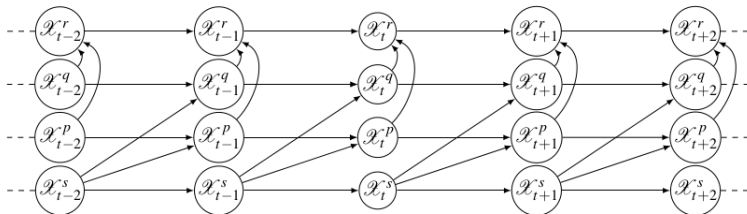
Window Causal Graph (b)



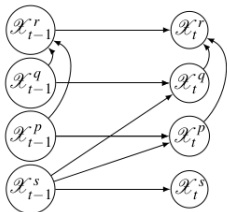
Summary Causal Graph (c)

Extracted from Assaad et al. 2022

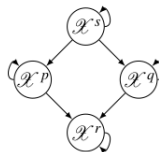
Background



Full Time Causal Graph (a)

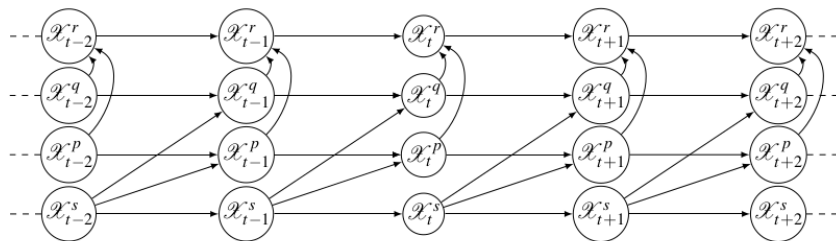


Window Causal Graph (b)



Summary Causal Graph (c)

Background



Full Time Causal Graph (a)

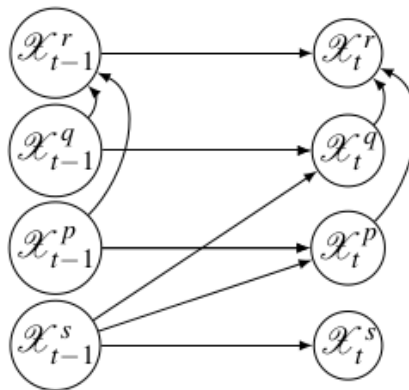
Background

Usually not possible to infer general full time causal graphs. A common assumption :

Consistency Throughout Time (Runge, 2018)

A causal graph $G = (V, E)$ for a multivariate time series X is said to be consistent throughout time if all the causal relationships remain constant in direction throughout time.

Background

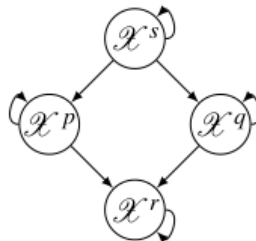


Window Causal Graph (b)

Background

- The window causal graph is a representation of the causal graph through a time window
- The size of the windows equals the maximum lag relating time series in the full time causal graph.

Background



Summary Causal Graph (c)

- In practice what practitioners know is the summary causal graph
- Much easier to estimate but less informative

Granger causality

Assume that we remove instantaneous effect. Then, the summary graph encodes the so called Granger causality

Justification of Granger causality

- Consider an SCM without instantaneous effects

$$X_t^{(j)} = f^{(j)}((PA_\tau^{(j)})_{t-\tau}, \dots, (PA_0^{(j)})_t, \varepsilon_t^{(j)})$$

- Assume that the induced joint distribution is faithful with respect to the corresponding full time graph
- Then the summary graph has an arrow from $X^{(j)}$ to $X^{(k)}$ if and only if

$$X_t^{(k)} \not\perp\!\!\!\perp X_{past(t)}^{(j)} | X_{past(t)}^{(-j)}$$

Granger causality

Test Granger causality?

- Assumption : multivariate linear time series
- Two models

$$X_t^{(k)} = a_{k,0} + \sum_{i=1}^{\tau} a_{k,i} X_{t-i}^{(k)} + \varepsilon_t^{(k)} \text{ (Mres)}$$

and

$$X_t^{(k)} = a_{k,0} + \sum_{i=1}^{\tau} a_{k,i} X_{t-i}^{(k)} + \sum_{i=1}^{\tau} a_{j,i} X_{t-i}^{(j)} + \varepsilon_t^{(k)} \text{ (Mfull)}$$

- If the full model is significantly more accurate than the restricted model, one concludes that $X^{(j)}$ Granger-causes $X^{(k)}$

Granger causality

Test Granger causality?

- Statistical test such as the F -test can be used to determine whether the full model is significantly better than the restricted one (null hypothesis: $X^{(j)}$ does not Granger-cause $X^{(k)}$)
- Optimal lag τ estimated using an information criterion, as AIC (Akaike information criterion) or BIC

Granger causality

input X a d -dimensional time series, $\tau_{\max} \in \mathbb{N}$ optimal lag
initialisation Form an empty graph $\mathcal{G}$ with d nodes V
Standardize data and check if it is covariance stationary
for $X^q \in V$ **do**
 Fit M_{res} and compute its residuals
 for $X^p \in V \setminus \{X^q\}$ **do**
 Fit M_{full} and compute its residuals
 Compare M_{res} and M_{full}
 if null hypothesis rejected **then** add $X^p \rightarrow X^q$ to $\mathcal{G}$
return $\mathcal{G}$

Information Theory approaches

Background : entropy

- Entropy of a discrete variable X

$$H(X) = - \sum_{x \in X} P_X(x) \log P_X(x) .$$

Measures amount of information (or uncertainty) in a random variable

- Conditional entropy

$$H(Y|X) = H(X, Y) - H(X).$$

It can be interpreted as the uncertainty about Y when X is known

Information Theory approaches

Transfer entropy between time series

- Transfer entropy from (X_t) to (Y_t) is the amount of uncertainty reduced in future values of Y by knowing the past values of X given past values of Y .
- Defined as

$$T_{X \rightarrow Y} = H(Y_t \mid Y_{t-1:t-L}) - H(Y_t \mid Y_{t-1:t-L}, X_{t-1:t-L}),$$

- Transfer entropy reduces to Granger causality for vector auto-regressive processes.
- Can be used also to test causality

Causal discovery

- Constraints based approach: PCMCI to discover a window causal graph
- Constraints based approach: PCGCE to discover an extended summary causal graph
- Causal discovery: noise based. VarLINGAM. Uses Lasso

The PCMC algorithm (Runge et al., 2019)

- One can adapt the PC algorithm to time series : PCMC algorithm
- Adapted versions of non-temporal causal graph discovery algorithms and consists of two steps procedure:
 - ▶ First step: use statistical tests in order to establish conditional dependence relationship among the variable in a model.
 - ▶ Second step: Constraint the types of causal relationships that exists among the model variables- orientation and false positive reduction

The PCMCI algorithm (Runge et al., 2019)

- Consider multivariate time series with

$$X_t^{(k)} = f^{(k)}(PA(X_t^{(j)}), \varepsilon_t^{(j)})$$

- Adapted version of non-temporal causal graph discovery: PC algorithm.
 - PC : reconstruct the graph by finding dependencies and parents -Markov set discovery algorithm PC1
 - MCI : test all pairs conditioned on the parents : False positive reduction

The PCMCI algorithm (Runge et al., 2019)

PC: identify the parents

- Initialize the parents:

$$PA(X_t^{(k)}) = (X_{t-1}, X_{t-2}, \dots, X_{t-\tau}).$$

- Unconditional test: remove $X_{t-\tau}^{(j)}$ from $PA(X_t^{(k)})$ if the test statistic $X_{t-\tau}$ ind from $X_t^{(k)}$ is rejected
- conditional test: sort parents by stat. value, and remove parents if the test statistic $X_{t-\tau}^{(j)}$ independent from $X_t^{(k)}$ conditionally to S is rejected where S is the p strongest parents

The PCMCI algorithm (Runge et al., 2019)

MCI: false reduction-orientation phase

Check $X_{t-\tau}^{(j)} \rightarrow X_t^{(k)}$: test statistics

$X_{t-\tau}^{(j)}$ ind from $X_t^{(k)} | PA(X_t^{(i)}) \setminus X_{t-\tau}^{(j)}, PA(X_{t-\tau}^{(j)})$

Causal inference

- Intervention problem
- BackDoorCriterion valid on the Full Time Causal Graphs
- Extension to summary graphs?
- Necessity to introduce a notion of abstract graphs